

Applied Agentic AI for PMs/TPMs

Capstone Project

PRD Genie: AI-Powered Product Documentation Assistant

Company Profile

NeuronForge Technologies is a fast-growing technology startup focused on building cutting-edge AI products for enterprise clients. The company’s mission is to empower organizations with intelligent automation solutions.

Your Role

You have recently joined NeuronForge as an AI PM/TPM within the Product and Innovation team. You are tasked with improving the requirement assessment process, automating PRD creation, improving identification of Epics/Features/User Stories, and increasing PMs/TPMs’ productivity.

Problem Statement

NeuronForge has identified a significant bottleneck in its product management process:

- PMs/TPMs spend a disproportionate amount of time gathering scope and requirements from meetings
- Translating discussions into structured PRDs, epics, and user stories is highly manual and iterative
- Different PMs/TPMs produce PRDs in different formats, making engineering estimation difficult
- Meeting transcripts and notes sit in documents nobody revisits — valuable requirements get lost
- This leads to delays, inconsistent tracking, and slower time-to-market

You are tasked with building PRD Genie — an AI-powered documentation assistant that ingests meeting transcripts, generates structured PRDs, and breaks them into epics and user stories.

Curriculum Concepts You Will Apply

Concept	Mandatory / Extended	How It Applies in PRD Genie
Multi-Agent System Design	Mandatory	Design agents for extraction, PRD generation, and story breakdown
Orchestration Pattern	Mandatory	Select and document sequential pipeline for transcript → requirements → PRD → stories
Prompt Engineering	Mandatory	Write prompts that distinguish stated requirements from assumptions
Data Grounding	Mandatory	Ground PRD content on actual transcript/notes, not LLM general knowledge
Evaluation & Observability	Mandatory	Connect to tracing tool, run test dataset, track hallucination

Fine-Tuning	Extended (optional)	Train a model to better extract requirements from messy transcripts
RAG / Knowledge Retrieval	Extended (optional)	Use PRD template and past PRDs as retrieval context
Human-in-the-Loop	Extended (optional)	PM/TPM reviews extracted requirements before PRD generation

Orchestration Pattern Requirements

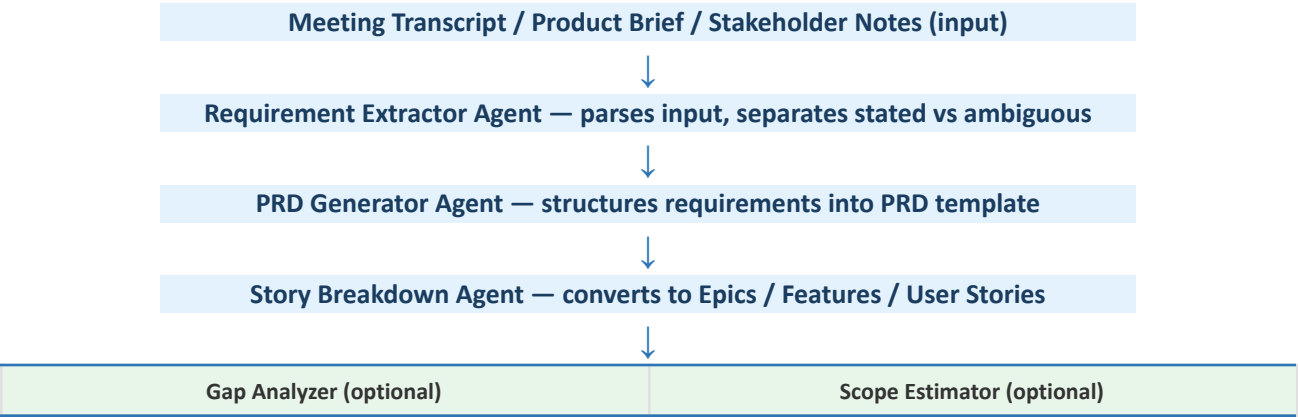
Your submission must document which orchestration pattern you selected and why.

Pattern	When to Use
Sequential Pipeline	Fixed-order processing: extract → generate → breakdown.
Router / Dispatcher	Route different input types (transcript vs brief vs notes) to specialized parsers.
Hierarchical	Decompose complex multi-document inputs into subtasks.

Recommendation for PRD Genie:
A Sequential Pipeline works well because every input follows the same stages: extract requirements → generate PRD → break into stories. If you add Gap Analysis or Scope Estimation, they can run in parallel after the Story Breakdown step.

Suggested Architecture Overview

Reference architecture. Design your own and justify your decisions.



Possible Agents

Agent	Role
Requirement Extractor	Parses transcripts/notes, extracts requirements, constraints, stakeholders, deadlines. Flags ambiguous items.
PRD Generator	Takes extracted requirements, produces structured PRD following a template
Story Breakdown	Converts PRD features into Epics and User Stories with priority suggestions
Gap Analyzer	Identifies missing info, produces clarification questions for stakeholders
Scope Estimator	Assigns T-shirt size estimates to user stories based on complexity signals

What PRD Genie Should Be Able to Do

Core Capabilities (Required)

Requirement Extraction: Ingest transcripts/notes, extract feature requests, requirements, constraints, assumptions, stakeholders, deadlines. Separate stated from ambiguous.

PRD Generation: Structured PRD including: overview, goals, user personas, feature list, acceptance criteria, NFRs, assumptions, open questions.

Epic and User Story Breakdown: Epics, Features, User Stories in standard format with priority (Must Have / Should Have / Nice to Have).

Extended Capabilities (Choose at least one)

Gap Analysis: Identify missing/ambiguous info, produce clarification questions.

Scope Estimator: T-shirt size (S/M/L/XL) for each story. Flag underspecified stories.

Version Comparison: Compare two transcript versions, identify new/modified/dropped requirements.

Evaluation Requirement

Connect your PRD Genie workflow(s) to any evaluation/observability tool. Run the baseline test dataset below and document results.

Baseline Test Dataset for Core Capabilities

Run each input. Check that outputs contain the expected keywords/elements.

ID	Test Input	Type	Expected Output Must Contain
T1	(Detailed transcript) 'The user should be able to filter reports by date range, category, and status. Results must load in under 2 seconds. PM: Sarah. Deadline: Q3.'	Detailed	Must extract: filter by date/category/status, 2-second load time, PM Sarah, Q3 deadline. Must NOT add requirements not in transcript.
T2	(Vague brief) 'We need better reporting. Something like what Competitor X has.'	Vague	Must flag as ambiguous. Must list missing info (what metrics, what format, what users). Must NOT invent specific requirements.
T3	(Contradictory) 'The dashboard should auto-refresh every 5 seconds.' Later: 'Performance is critical, minimize API calls.'	Contradictory	Must identify the contradiction (frequent refresh vs minimize calls). Must flag for stakeholder resolution.
T4	(Feature request with acceptance criteria) 'Users need to export reports as PDF and CSV. PDF must include company logo. CSV must preserve formulas.'	Detailed with AC	Must extract acceptance criteria verbatim. Must generate User Stories with these as criteria. Must NOT add extra criteria.
T5	(Incomplete notes) 'Discussed dashboard... John mentioned something about real-time... budget TBD...'	Incomplete	Must flag as insufficient. Must list what is missing (what dashboard, what real-time, what budget). Must NOT fill in gaps with assumptions.
T6	(Multi-stakeholder) 'Engineering wants microservices. Design wants single-page app. PM wants it shipped by March.'	Multiple viewpoints	Must capture all three viewpoints. Must identify tensions. Must NOT favor one over others.
T7	(Technical requirements) 'API must support 10,000 concurrent users. Response time < 200ms at p95. Must integrate with Salesforce REST API v52.'	Technical	Must extract exact numbers (10K users, 200ms p95, Salesforce v52). Must classify as NFRs. Must NOT round or modify the numbers.
T8	(User persona heavy) 'Admins need bulk user management. End users need a simplified view. Auditors need read-only access with full history.'	Persona-based	Must identify 3 personas (Admin, End User, Auditor). Must create separate stories for each. Must NOT merge into generic user stories.
T9	(Empty/minimal) 'Meeting happened. Notes: none.'	Edge case	Must flag as no requirements extractable. Must NOT generate a PRD from nothing.
T10	(Feature with dependencies) 'SSO login requires the new auth service which is being built by Team Alpha. ETA unknown.'	Dependency	Must extract the feature (SSO). Must flag dependency on auth service / Team Alpha. Must flag unknown ETA as a risk.

T11	Generate a PRD from T1 extraction.	PRD generation	Must follow template structure. Must include all sections. Must only contain requirements from T1.
T12	Break T11 PRD into user stories.	Story breakdown	Must produce Epics and User Stories in 'As a [user]' format. Must include priority suggestions.

Fine-Tuning (Optional):

Fine-tune the Requirement Extractor to better distinguish stated requirements from assumptions. Not required.

Inputs Provided

Input File	Description
sample_meeting_transcripts.txt	3-5 meeting transcript excerpts of varying quality
sample_product_brief.txt	1-page product brief for a hypothetical feature
prd_template.md	Standard PRD template showing expected output structure
stakeholder_notes.txt	Raw notes from stakeholder interviews with incomplete information
eval_prdgenie_inputs.txt	Baseline test inputs (the table above in text format)

Tools and Technologies

Category	What You Need	Examples
Workflow/Agent Platform	Low-code or no-code platform	n8n, LangFlow, or any
LLM	Reasoning, extraction, generation	OpenAI, Claude, Gemini
Document Ingestion	Read/parse transcripts, notes	Built-in nodes, HTTP, code
Observability	Monitor traces, track quality	LangWatch, Langfuse, any
Output/Export	Structure outputs	Markdown, Google Docs, Notion, text
Authentication	Secure access	OAuth2, API keys

Assignments

Q1: Ideation (15 points)

Identify pain points in NeuronForge's PM workflow. For each: manual step, agent solution, inputs/outputs, risks.

Q2: Program Charter (15 points)

Draft program charter for PRD Genie: vision, objectives, scope, success criteria, timeline, risks, stakeholders, rollout plan.

Q3: Build PRD Genie (45 points)

Design/Architecture (diagram, orchestration pattern, tool rationale, agent decisions, human-in-the-loop), Implementation (core + extended, observability, baseline dataset testing), Cost/Evaluation Strategy. 1-2 page writeup.

Sub-criteria	Points
Design: architecture + orchestration + tool rationale	10
Core capabilities end-to-end	12
Extended capability	8
Observability connected	5
Baseline dataset tested with results	5
Cost analysis and evaluation strategy	5

Q4: Reflection (5 points)

1-page reflection: trace findings, improvement plan, risks of AI-generated PRDs, connection to evaluation skills.

Submission Guidelines

- Export workflow(s) as JSON, upload to GitHub with README
- Screenshots: canvas, PRD Genie in action, traces
- Architecture diagram, slide deck, 5-min demo video
- Baseline test results documented (all 12 inputs with outputs)
- Submit GitHub link along with documents